Nothing Here

Fast Quantum Algorithms

or

How we learned to put our pants on two legs at a time.



Dave Bacon

Institute for Quantum Information

California Institute of Technology

?

A sudden bold and unexpected question doth many times surprise a man and lay him open.

A prudent question is one-half of wisdom.



Sir Francis Bacon
(1561-1628)

Iway amway Akespeareshay!



William Shakespeare
(1568-1623)

"small Latin, less Greek" ?

WarNING

This Talk Under Constant Acceleration

DB and CBSSS assume no responsibility for injuries sustained while zoning out.

Quantum Computers Can Do Amazing Things!

THIS TALK

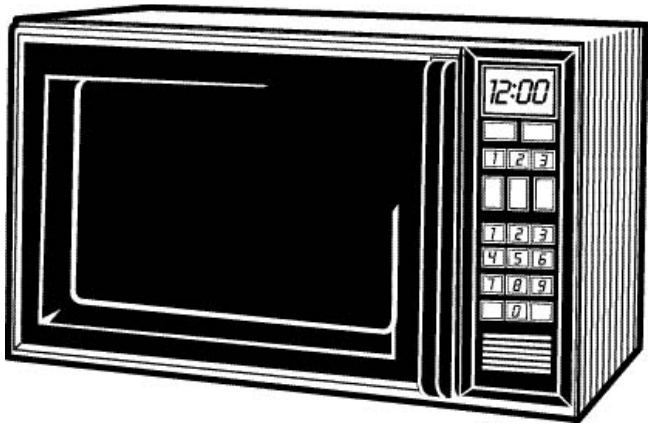
Understanding what makes quantum evolution *different*

How quantum evolution can used to do something cool.

How quantum evolution can be used to exponentially speed up an oracle problem over classical deterministic algorithms.

How quantum evolution can be used to exponentially speed up an oracle problem over classical probabilistic algorithms.

Randomizing Microwave



Mystery Markov Microwave

Digital Coffee
(Not Java!)



Scalding Hot

H



Freezing Cold

C



Marko

The true method of knowledge is experiment. - William Blake 1788

V

- Run Experiments To Understand MMM Machine

If you put in C, 70% of the time you get H out
and 30% of the time you get C out
If you put in H, 80% of the time you get H out
and 20% of the time you get C out

- A nice little formalism

$$\mathbf{H} \longrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\mathbf{C} \longrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$M_{mm} = \begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}$$

$$\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix}$$

$$\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix}$$

columns sum to 1

$0 \leq$ matrix entry ≤ 1

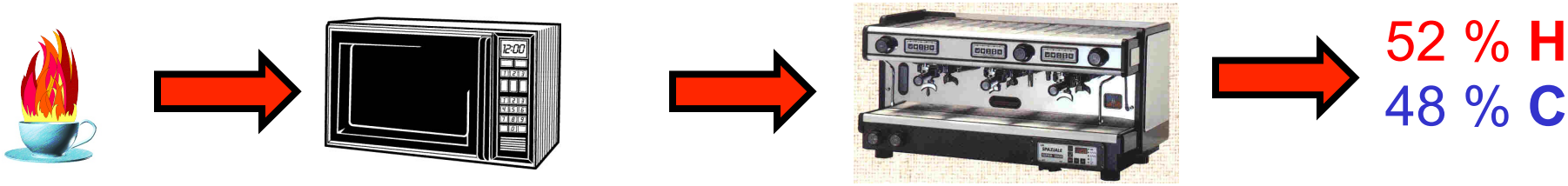
Markov Chains



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} \quad \begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}\begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0.78 \\ 0.22 \end{pmatrix}$$

or

$$\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} = \begin{pmatrix} 0.78 & 0.77 \\ 0.22 & 0.23 \end{pmatrix} \quad \begin{pmatrix} 0.78 & 0.77 \\ 0.22 & 0.23 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.78 \\ 0.22 \end{pmatrix}$$



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} \quad \begin{pmatrix} 0.5 & 0.6 \\ 0.5 & 0.4 \end{pmatrix}\begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0.52 \\ 0.48 \end{pmatrix}$$

or

$$\begin{pmatrix} 0.5 & 0.6 \\ 0.5 & 0.4 \end{pmatrix}\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} = \begin{pmatrix} 0.52 & 0.53 \\ 0.48 & 0.47 \end{pmatrix} \quad \begin{pmatrix} 0.52 & 0.53 \\ 0.48 & 0.47 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.52 \\ 0.48 \end{pmatrix}$$

Quantum Microwave

Quantum Digital Coffee



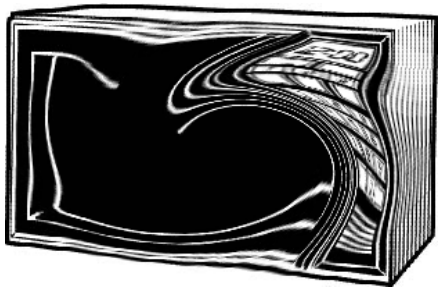
Scalding Hot

H



Freezing Cold

C



Quantum Microwave (QM)

What are the rules for the Quantum Microwave?

The Amplitude Attitude

H **C**

qubit

Complex numbers $(a+bi)$ $\mathbb{C}$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$|0\rangle$ $|1\rangle$ $\alpha|0\rangle + \beta|1\rangle$

If we measure the state then

with probability $|\alpha|^2$ we get H

with probability $|\beta|^2$ we get C

$$|\alpha|^2 + |\beta|^2 = 1$$

For Our Purposes

Real numbers $\mathbb{R}$

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$\alpha|0\rangle + \beta|1\rangle$

$-1 \leq \alpha \leq 1$

$-1 \leq \beta \leq 1$

$\alpha^2 + \beta^2 = 1$

If we measure the state then

with probability α^2 we get H

with probability β^2 we get C

Expert Note: Quantum computation with real numbers is equivalent to quantum computation with complex numbers as a consequence of $SU(N) \subseteq SO(2N)$.

Unitary

$\neq$ COVENANT OF UNITARIAN UNIVERSALIST PAGANS

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \rightarrow \boxed{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}} \rightarrow \begin{pmatrix} \frac{\alpha+\beta}{\sqrt{2}} \\ \frac{\alpha-\beta}{\sqrt{2}} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \frac{\alpha+\beta}{\sqrt{2}} \\ \frac{\alpha-\beta}{\sqrt{2}} \end{pmatrix}$$

In general this matrix U must be unitary:

$$UU^\dagger = I$$

i.e. the rows U are orthogonal vectors.

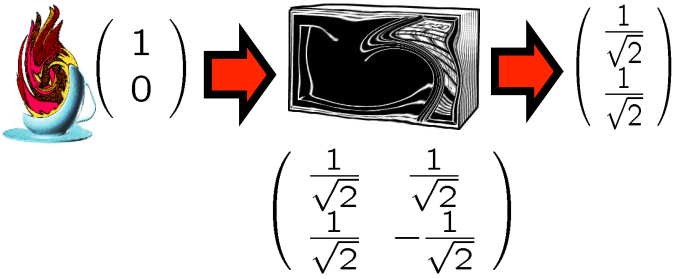
For our purposes (real numbers), this becomes

$$UU^T = I$$

Matrix entries will be between -1 and 1

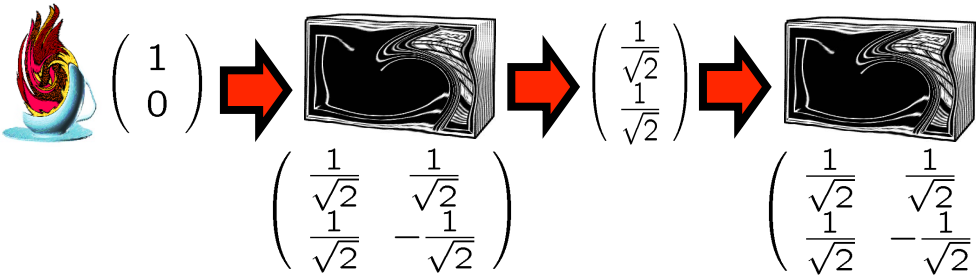
Interference





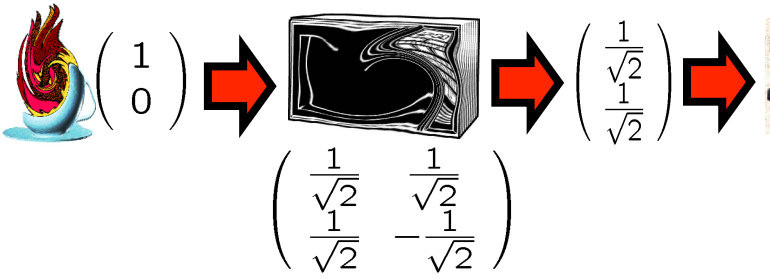
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

50 % **H**
50 % **C**



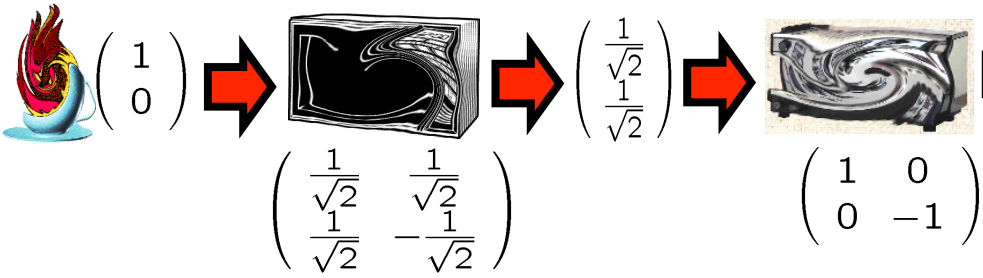
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

100 %
H
0 % **C**



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

50 % **H**
50 % **C**



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

0 % **H**
100 %
C

Deutsch's Problem



David Deutsch



Dr. Falcon

(1985)

two qubits

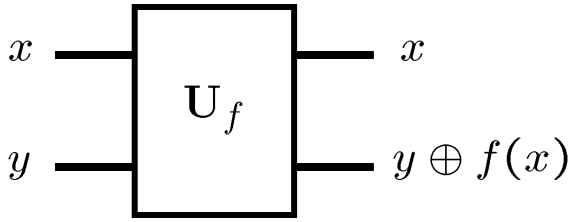
$$\alpha_0|00\rangle + \alpha_1|01\rangle + \alpha_2|10\rangle + \alpha_3|11\rangle$$

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix}$$



Delphi

Oracle



$x, y, f(x) \in \{0, 1\}$

Example $f(x) = x$:

$$U_f = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Four possible functions $f(x)$:

$f(x) = 0$ $f(x) = 1$ $f(x) = x$ $f(x) = \bar{x}$

Constant functions Balanced functions

Deutsch's Problem

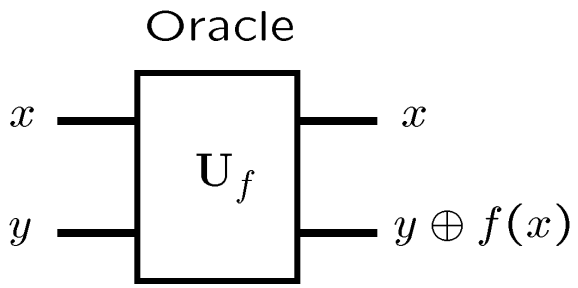
Determine whether f(x) is constant or balanced using as few queries to the oracle as possible.

Classical Deutsch

Four possible functions $f(x)$:

$f(x) = 0 \quad f(x) = 1 \quad f(x) = x \quad f(x) = \bar{x}$

Constant functions Balanced functions



$x, y, f(x) \in \{0, 1\}$

Query input of x_0 and y_0 only gives information about $f(x_0)$.

Knowing $f(x_0)$ not enough to distinguish constant from balanced.

Classically we need to query the oracle <u>two</u> times to solve Deutsch's Problem

Quantum Deutsch

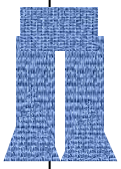


1. Query oracle with $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$
2. Producing the state $\frac{1}{2}\left(|0f(0)\rangle - |0\bar{f}(0)\rangle + |1f(1)\rangle - |1\bar{f}(1)\rangle\right)$

$f(x) = 0$	$f(x) = 1$	$f(x) = x$	$f(x) = \bar{x}$
$\frac{1}{2}(\|00\rangle - \|01\rangle + \|10\rangle - \|11\rangle)$	$\frac{1}{2}(\|01\rangle - \|00\rangle + \|11\rangle - \|10\rangle)$	$\frac{1}{2}(\|00\rangle - \|01\rangle + \|11\rangle - \|10\rangle)$	$\frac{1}{2}(\|01\rangle - \|00\rangle + \|10\rangle - \|11\rangle)$
$\frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}$	$\frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \end{pmatrix}$	$\frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$	$\frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix}$
Constant functions		Balanced functions	

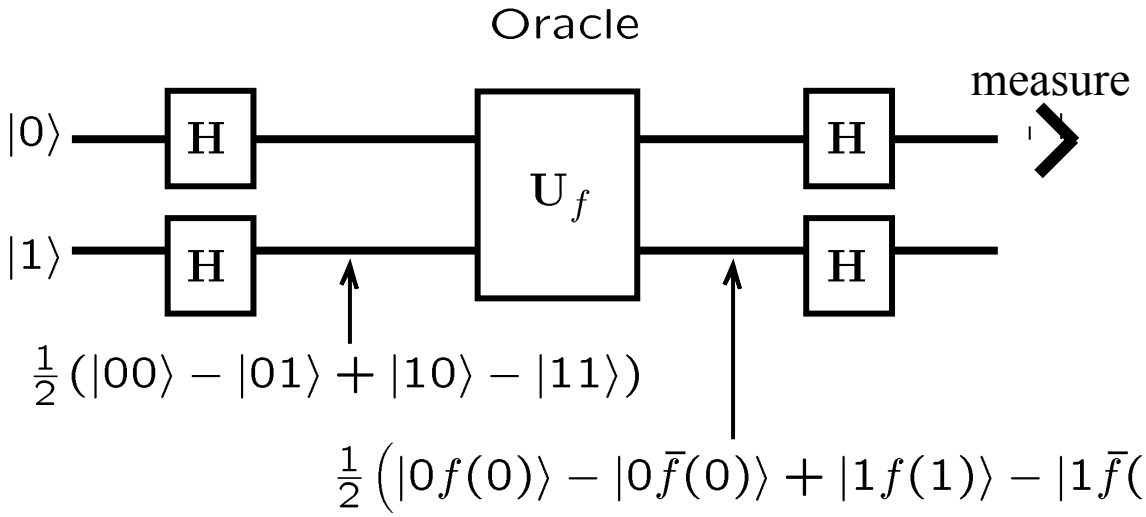
3. Apply the unitary transformation

$$\frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

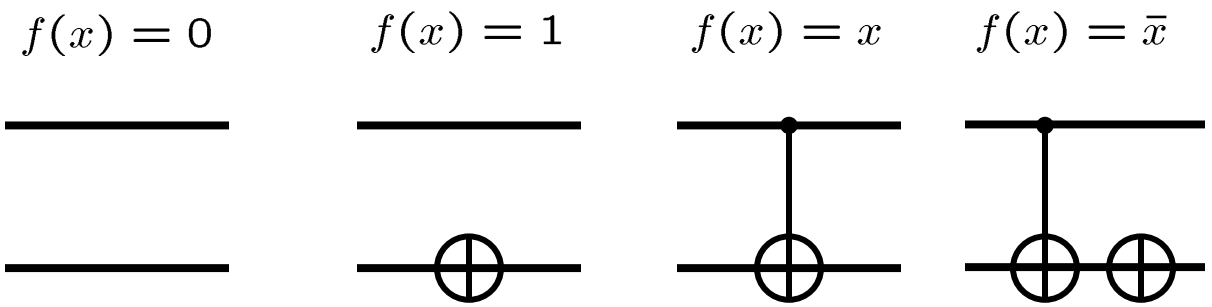


$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}$
100 %	100 %	100 %	100 %

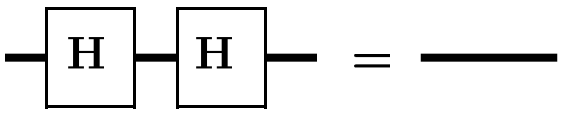
Deutsch Circuit



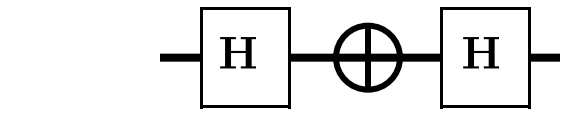
$$\mathbf{H} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \qquad \oplus = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



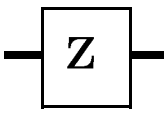
A Different View



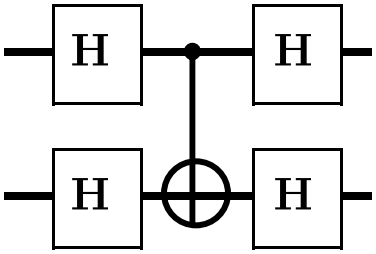
$=$



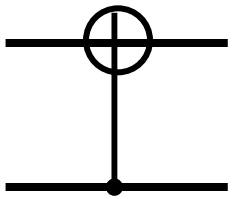
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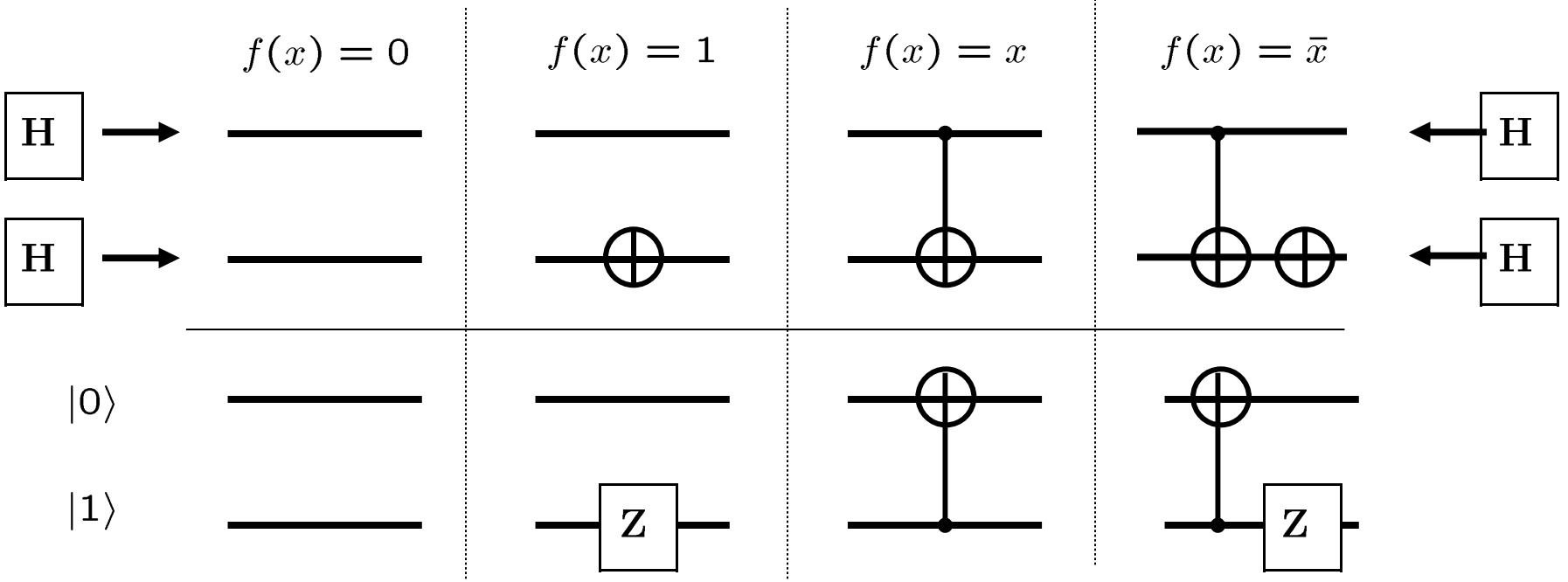


$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



$=$





Deutsch-Jozsa Problem

(1992)

Given a function $f : \{0,1\}^n \rightarrow \{0,1\}$

function maps n bit strings to a single bit

f is promised to be either

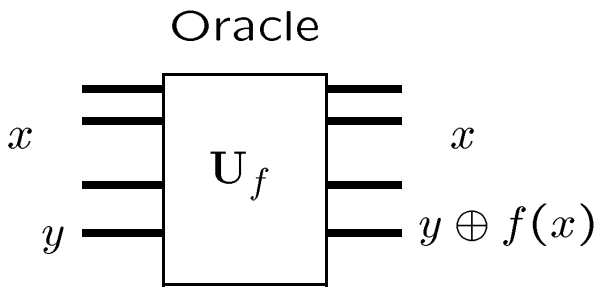
(A) f is constant.

$f(x) = b \ \forall x$

(B) f is balanced.

$f(x) = 1$ if $x \in \mathcal{S}$ otherwise $f(x) = 0$

$\mathcal{S}$ has 2^{n-1} elements.



$x \in \{0,1\}^n$

$y, f(x) \in \{0,1\}$

Deutsch-Jozsa Problem

Determine whether f(x) is constant or balanced using as few queries
to the oracle as possible.

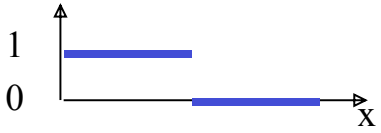
Classical DJ





(A) f is constant.

$f(x) = b \ \forall x$



(B) f is balanced.

$f(x) = 1$ if $x \in \mathcal{S}$ otherwise $f(x) = 0$

$\mathcal{S}$ has 2^{n-1} elements.

If we never want to be wrong:

Worst case we need $2^{n-1} + 1$ querries

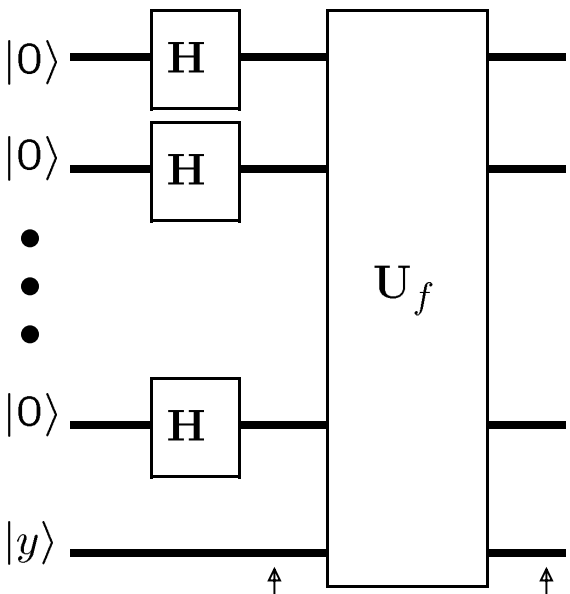
If we allow a failure probability of ϵ, then

Randomly choose k different x_k to query.

$k = O\left(log\left(\frac{1}{\epsilon}\right)\right)$ (question on website!)

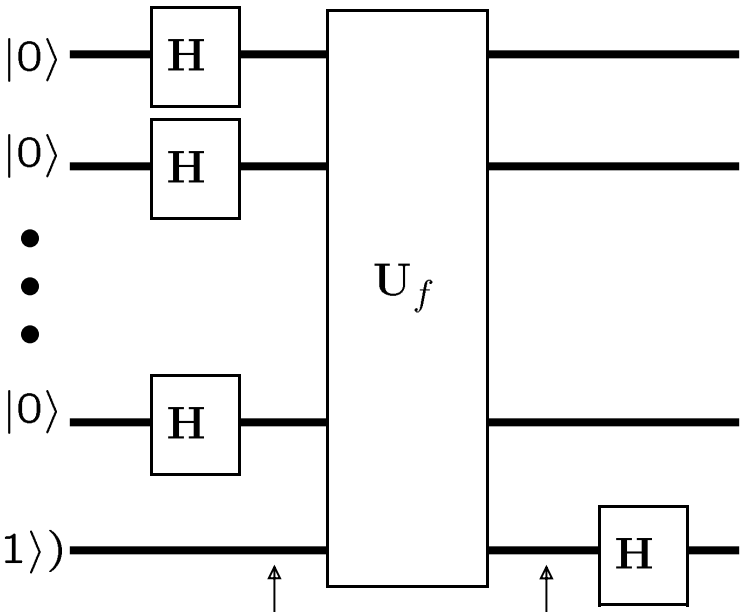
Determinstically hard, probabilistically easy.

Quantum DJ



$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|y\rangle \quad \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|y \bigoplus f(x)\rangle$$

$$|x\rangle = |x_n x_{n-1} \ldots x_1\rangle$$



$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|0\rangle - |1\rangle\right)$$

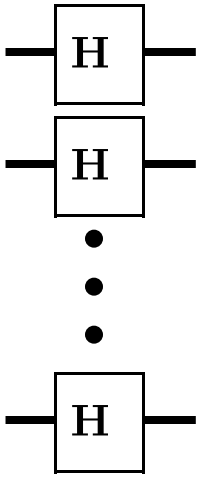
$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|f(x)\rangle - |\bar{f}(x)\rangle\right)$$

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle|0\rangle$$

Quantum DJ

Information about $f(x)$ is in the phase

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle|0\rangle$$



$= \mathbf{H}^{\otimes n}$

$$\mathbf{H}^{\otimes n}|x_n x_{n-1} \ldots x_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle+(-1)^{x_n}|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle+(-1)^{x_{n-1}}|1\rangle)\cdots\frac{1}{\sqrt{2}}(|0\rangle+(-1)^{x_1}|1\rangle)$$

$$= \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x\cdot y}|y\rangle \qquad x \cdot y = x_n y_n + x_{n-1} y_{n-1} + \cdots + x_1 y_1 \mod 2$$

All matrix elements of $\mathbf{H}^{\otimes n}$ are $\pm\frac{1}{\sqrt{2^n}}$

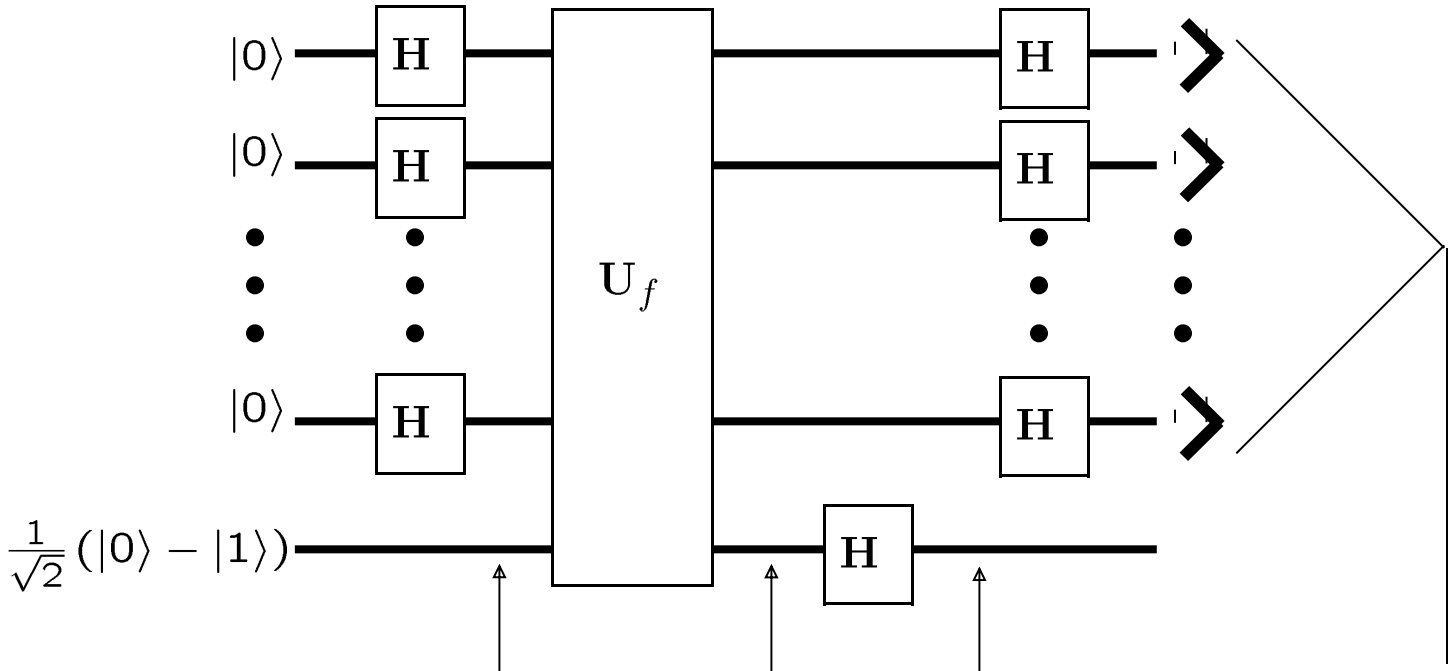
First row of $\mathbf{H}^{\otimes n}$ has all elements $+\frac{1}{\sqrt{2^n}}$

Other rows have equal number $\pm\frac{1}{\sqrt{2^n}}$ elements.

If $f(x)$ is constant, applying $\mathbf{H}^{\otimes n}$ to $\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle$ always gives $|0\rangle$.

If $f(x)$ is balanced, applying $\mathbf{H}^{\otimes n}$ to $\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle$ always gives a superposition without $|0\rangle$.

Full Quantum DJ



$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|0\rangle - |1\rangle\right)$$

$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|f(x)\rangle - |\bar{f}(x)\rangle\right)$$

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle|0\rangle$$

If all bits are 0, then f is constant.

If all bits are not 0, then f is balanced.

Solves DJ with a SINGLE query vs 2n-1+1 classical deterministic!!!!!!!!

Simon's Problem

(is that no one does what "Simon says"?) (1994)

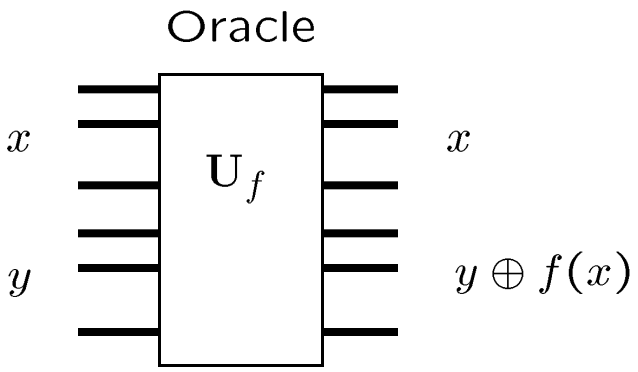
Given a function $f : \{0,1\}^n \rightarrow \{0,1\}^n$

function maps n bit strings to a n bit strings

f is promised to satisfy either

(A) f is distinct on all inputs.

$$f(x) \neq f(y) \quad \forall x \neq y$$



(B) f is distinct up to an XOR mask s

$$f(x) = f(x \oplus s)\ \forall x \qquad (s \text{ is unkown and } \neq 0)$$

$$f(x) \neq f(x \oplus t)\ \forall t \neq s$$

Simon's Problem

Determine whether f(x) has is distinct on an XOR mask or distinct
on all inputs using the fewest queries of the oracle. (Find s)

Classical Simon

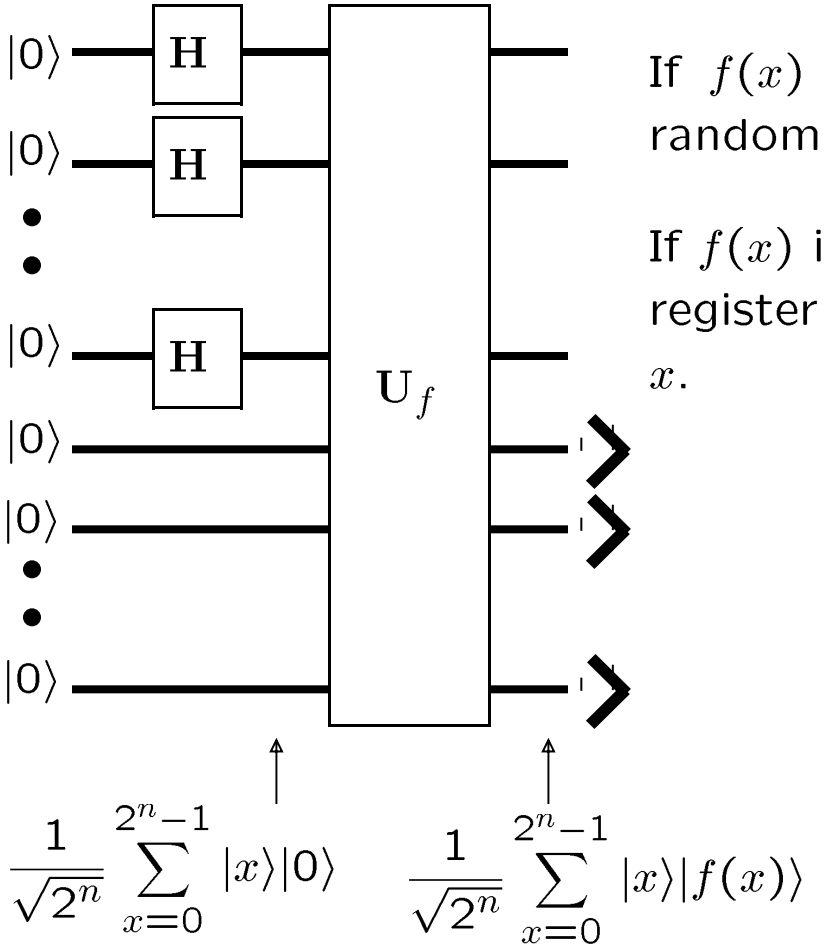


Now both determinsitcally and probabilistically an exponential number of queries is required.

Left as an exercise for those who are asleep.

Make sure to tell the sleepers that they missed a cameo appearance by Robin Williams.

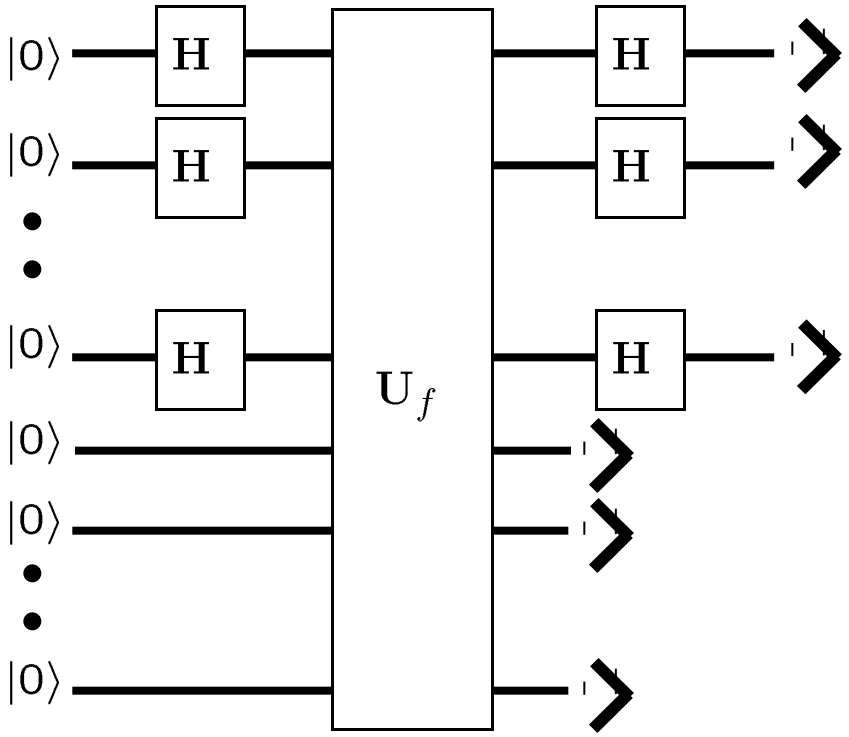
Quantum Simon



If $f(x)$ is distinct, then first register will be random $|x\rangle$.

If $f(x)$ is distinct on an XOR map s, then first register will be $\frac{1}{\sqrt{2}}(|x\rangle + |x \oplus s\rangle)$ for a random x.

Quantum Simon



Quantum Simon

$$\mathbf{H}^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x\cdot y}|y\rangle$$ ← Equal amplitude and uniform over all y

$$\mathbf{H}^{\otimes n}\frac{1}{\sqrt{2}}(|x\rangle + |x\oplus s\rangle) = \frac{1}{\sqrt{2^{n+1}}} \sum_{y=0}^{2^n-1} \left((-1)^{x\cdot y}|y\rangle + (-1)^{(x\oplus s)\cdot y}|y\rangle\right)$$

$$= \frac{1}{\sqrt{2^{n+1}}} \sum_{y=0}^{2^n-1} (-1)^{x\cdot y}(1 + (-1)^{s\cdot y})|y\rangle \quad (1)$$

↑

Equal amplitude and uniform over all y such that $y \cdot s = 0$.

If we get $n-1$ linearly independent y's such that $y \cdot s = 0$ we can determine s.

Claim: with probability at most $\frac{1}{4}$, $n-1$ random vectors y such that $y \cdot s = 0$ are linearly independent.

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Shor Type Algorithms

1) Deutsch's algorithm demonstrates task quantum computer can perform in one shot that classically takes two shots.

1992 Deutsch-Jozsa algorithm demonstrates an exponential separation between classical deterministic and quantum algorithms.

1993 Bernstein-Vazirani demonstrates a superpolynomial algorithm separation between probabilistic and quantum algorithms.

1994 Simon's algorithm demonstrates an exponential separation between probabilistic and quantum algorithms.

1994 Shor's algorithm demonstrates that quantum computers can efficiently factor numbers.